

HW 12

9.2

Let X be an exponential random variable with parameter $\lambda = \frac{1}{2}$

a

Use Markov's inequality to find upper bound for $P(X > 6)$.

✓ Answer ✓

$$E[X] = 2$$

$$\frac{2}{6} = 0.33$$

b

Use Chebyshev's inequality to find an upper bound for $P(X > 6)$.

✓ Answer

$$\text{Var}(X) = 4$$

$$\frac{4}{16} = \frac{1}{4} = 0.25$$

c

Explicitly compute the probability above and compare with the upper bounds you derived.

✓ Answer

$$-e^{-x/2} \Big|_6^{\infty} = e^{-3}$$

$$\approx 5\%$$

9.7

A car insurance company has 2500 policy holders. The expected claim paid to a policy holder during a year is \$1000 with a standard deviation of \$900. What premium should the company charge each policy holder to assure that with probability 0.999, the premium income will cover the cost of the claims? Compute the answer both with Chebyshev's inequality and with the CLT.

✓ Answer

$P \sim N(1000, 900)$ by CLT.

$$P(S_{P_n} > \text{price}) = 0.999 \implies x = 3.09$$

$$S_{P_n} \sim N(2500 * 1000, 900 * \sqrt{2500})$$

$$2500 * 1000 + 3.09 * 900 * \sqrt{2500} = 2639050$$

$$\implies \text{price} > 1055.62$$

By Chebyshev's,

$$0.001 = \frac{900 * \sqrt{2500}}{\text{price}^2}$$

$$\text{extra price} > 1423024.9470757707$$

$$\text{price} > 1569.2099788303083$$

9.12

Let X be a $\text{Poisson}(2)$ random variable, and Y an independent $\text{Unif}(0, 1)$ random variable.

a

Use Markov's inequality to bound $P(X + Y > 10)$.

✓ **Answer**

X and Y are independent, $E[X + Y] = 2 + 0.5 = 2.5$

$$\text{Var}(X + Y) = 2 + \frac{1}{12} = \frac{25}{12}$$

Using Markov,

$$P(X + Y > 10) = \frac{2.5}{10} = 0.25$$

b

Use Chebychev's inequality to bound $P(X + Y > 10)$.

✓ **Answer**

$$\frac{\frac{25}{12}}{7.5^2} = 0.037$$

10.4

Let N have a $\text{Bin}(100, \frac{1}{4})$ distribution. Given that $N = n$, flip a fair coin n times, and let X be the number of heads observed.

a

Find the conditional distribution of X given that $N = n$. Be precise about the possible values and a proper condition probability mass function.

✓ Answer

$$X \sim \text{Binom}(n, 0.5)$$

$$P(X = x | N = n) = \binom{n}{x} \left(\frac{1}{2}\right)^n$$

The support is $[0, n] \in \mathbb{Z}$

b

Find $E[X|N = n]$ for all the values of n that it is defined.

✓ Answer

$$E[\text{Binom}(n, 0.5)] = 0.5n, n \in [0, 100] \in \mathbb{Z}$$

c

Use part b to compute $E[X]$

✓ Answer

$$E[X] = E\left[\frac{N}{2}\right] = \frac{1}{2}E[N] = \frac{25}{2} = 12.5$$

10.6

Let the joint density function of (X, Y) be

$$f(x, y) = \frac{x+y}{4}, 0 < x < y < 2.$$

a

Find $f_{X|Y}(x|y)$

✓ Answer

$$f_Y(y) = \int_0^y \frac{x+y}{4} dx = \frac{3y^2}{8}$$

$$f_{X|Y}(x|y) = \frac{\frac{x+y}{4}}{\frac{3y^2}{8}} = \frac{2(x+y)}{3y^2}, 0 < x < y < 2$$

b

Compute the conditional probabilities $P(X < \frac{1}{2} | Y = 1)$ and $P(X < \frac{3}{2} | Y = 1)$.

✓ Answer

$$P\left(X < \frac{1}{2} | Y = 1\right) = \int_0^{1/2} \frac{2}{3}(x+1) = \frac{2}{3} \frac{5}{8} = \frac{5}{12}$$

$$P\left(X < \frac{3}{2} | Y = 1\right) = 1 \text{ because } X < Y$$

c

Compute the conditional expectation $E[X^2 | Y = y]$. Check the averaging identity

$$\int_{-\infty}^{\infty} E[X^2 | Y = y] f_Y(y) dy = E[X^2]$$

✓ **Answer**

$$= \int_0^y \frac{2}{3} x^2 \frac{x+y}{y^2} dx = \frac{2}{3} y^2 \left(\frac{y^4}{4} + \frac{y^4}{3} \right) = \frac{7y^2}{18}$$

$$\int_0^2 \frac{7y^4}{48} dy = \frac{14}{15}$$

$$\int_0^2 \int_0^y x^2 \frac{x+y}{4} dx dy = \int_0^2 \frac{7y^4}{48} dy = \frac{14}{15}$$

10.15

Consider a large insurance company with two types of policies: policy A and policy B. Suppose that the number of claims the company sees in a given day has a Poisson distribution with a parameter of 12. Suppose further that a randomly selected claim is from a type A policy with probability $\frac{1}{4}$.

a

Find the probability that the company will receive at least 5 claims from A policies tomorrow.

✓ **Answer**

$$A \sim \text{Poisson}\left(\frac{12}{4} = 3\right)$$

$$B \sim \text{Poisson}\left(\frac{12(3)}{4} = 9\right)$$

$$P(A \geq 5) = 0.18474$$

b

Find the probability that the company will receive at least 5 claims from B policies tomorrow.

✓ **Answer**

$$P(B \geq 9) = 0.54434$$

c

Find the probability that the company will receive at least 10 total claims tomorrow.

✓ **Answer**

$$T \sim \text{Poisson}(12)$$

$$P(T \geq 10) = 0.75761$$

10.38

We generate a random variable $Y \sim \text{Exp}(1)$. Then we generate another exponential random variable X whose parameter is the value Y just observed.

a

Find the joint density function of (X, Y) .

✓ **Answer**

$$Y \sim \text{Exp}(1)$$

$$X \sim \text{Exp}(Y)$$

$$f_{X,Y}(x, y) = f_{X|Y}(x|y)f_Y(y) = e^{-y}(ye^{-yx}) = ye^{-(x+1)y}, x > 0, y > 0$$

b

Suppose we observe $X = x$, How is Y now distributed? Identify the distribution by name.

✓ **Answer**

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} \propto f_{X,Y}(x, y)$$

$$\sim \text{Gamma}(2, x + 1)$$